LCSC Cable - quotation dossier

RFQ-CABLE-01 | 09 November 2026 | 10 of EACH listed assembly

QUOTATION ONLY - NOT RELEASED FOR PRODUCTION

13 quotation lines /130 assemblies if all lines were selected:8 internal definitions (80pcs, including optionalTMP117) and5 provisional external/service definitions (50pcs). Price each separately. No order, sample build or supplier contact has been authorized by this package.

ID	Assembly	Qty	Quotation scope
H01	A-B POWER	10	INTERNAL
H02	A-B I2C	10	INTERNAL
H03	A-B CTRL	10	INTERNAL
H04	LCD I2C	10	INTERNAL
H05	Encoder completed subassembly	10	INTERNAL
H06	TMP117 direct XH-SH option	10	INTERNAL OPTION
H07	BH1750 GY302 I2C	10	INTERNAL
H08	Internal probe panel socket subassembly	10	INTERNAL
H09	External jacketed NTC probe option	10	PROVISIONAL
H10	DC supply input cord	10	PROVISIONAL
H11	Heater output cord	10	PROVISIONAL
H12	Independent thermal cutoff lead set	10	PROVISIONAL
H13	SideUSB-C data extension	10	PROVISIONAL

Eight internal lines define connector families and electrical maps. Provisional linesH09-H13 need supplier proposals and purchaser application choices; do not treat their assumed lengths or undefined ends as manufacturing approval.

Request separate materials/labour/tooling/sample/shipping/tax costs and lead times. List exact suppliedMPNs and every deviation. Include the encoder and panel jack as finished workshop-soldered subassemblies or price consigned parts separately.

Common construction and approval requirements

Length datum

Flexible wire-path length between rear wire exits of connector housings, or solder joints/terminal shoulders for soldered ends. Includes fan-outs, branches and transitions along that path, excludes connector bodies and contacts. Supplier must state exact datum and physical tip-to-tip length on approval drawing; do not silently interpret as cut-wire length. 300mm is a quotation/service-loop proposal, actual routed fit unqualified.

Pin numbering and views

Every number is an electrical cavity number per manufacturer drawing, not a left-to-right photo index. Diagrams are logical wiring, NOT physical connector face views. Supplier shall provide both mating-face and wire-entry views with key/latch, cavity1, wire colours and drawing view direction before manufacture. Never mirror cavities based on looking at the other end.

Wire, jacket and labels

Flexible copper stranded wire, insulation at least80degC, ratings declared by supplier. Continuous flexible insulating common jacket preferred (PVC or documented equivalent); no two loose insulated wires as external probe cable. Nominal internal fan-out30mm each end unless individual line states otherwise; common jacket stops before crimps and latches. Supplier proposes jacket ID/OD, wire insulation OD, bend radius and temperature rating. Add labels at both ends, individual function labels on flying connectors, and retained heat-shrink at jacket transitions without stiffening latches. PET braid alone is not insulating common jacket. No rigid potting or excess adhesive on clips/connectors.

Parts and substitutions

Quote specified genuine JST/Harwin references first, with traceable sourcing. Unavailable parts may be proposed with exact manufacturer/MPN and drawings as separately identified alternatives requiring approval. No substitution by pitch/appearance alone; XH2.50mm, SH1.00mm and header2.54mm are distinct. Do not substitute low-insertion-force SXH-001T-P0.6N without approval.

Inspection

100percent continuity and unintended-short inspection per supplied map, including CTRL5/6 wires. Check conductor resistance and report method/results; specify acceptance limits using selected wire datasheets and length. Inspect contact retention, crimp conductor/insulation support, dimensions, labels and jacket anchoring; sampled crimp pull test to applicable connector/wire specification, report method and sample count. No unagreed high-voltage test across NTC/encoder/USB electronics. No solder-tinned stranded wire under screw terminals. First-article verification and purchaser approval drawing required before any production release.

Source drawings and references

https://lcsccable.com/faqs/ordering/submit-custom-cable-order-or-rfq

https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

https://www.jst-mfg.com/product/pdf/eng/eSH.pdf

https://www.harwin.com/products/M20-1060100

https://www.harwin.com/products/M20-1160042

https://learn.adafruit.com/introducing-adafruit-stemma-qt/technical-specs

https://learn.adafruit.com/adafruit-tmp117-high-accuracy-i2c-temperature-monitor/pinouts

https://www.gotronic.fr/art-encodeur-24-impulsions-ada377-37481.htm

https://www.taydaelectronics.com/datasheets/files/PJ-392.pdf

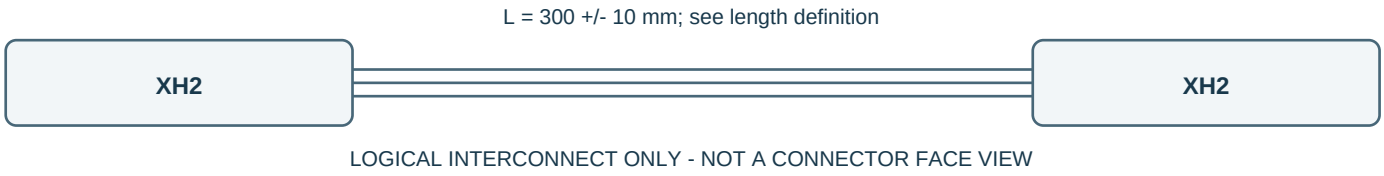
https://www.phoenixcontact.com/en-gb/products/printed-circuit-board-terminal-mkds-15-2-508-1715721

H01 | A-B POWER

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J1 - JST XHP-2; JST SXH-001T-P0.6

End B: B/J1 - JST XHP-2; JST SXH-001T-P0.6



Wire: 22AWG | Existing A-B quotation contract; physical service route pending

A cavity / function	B cavity / function	Signal / purpose	Colour
1	1	GND	black
2	2	+5V_AB	red

Jacket and finish

Common insulating flexible jacket; end fan-outs30mm nominal, pin1 black and pin2 red; label5V ONLY / NO HEATER CURRENT

Approval points / supplier response

Electrical map fixed; supplier drawing, OD/bend radius and real routed service slack to approve

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

H02 | A-B I2C

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J6 - JST XHP-3; JST SXH-001T-P0.6

End B: B/J6 - JST XHP-3; JST SXH-001T-P0.6

L = 300 +/- 10 mm; see length definition

LOGICAL INTERCONNECT ONLY - NOT A CONNECTOR FACE VIEW

Wire: 26AWG | Existing A-B quotation contract; physical service route pending

A cavity / function	B cavity / function	Signal / purpose	Colour
1	1	GND	black
2	2	I2C_SDA	blue
3	3	I2C_SCL	yellow

Jacket and finish

Common insulating flexible jacket; pin/function labels both ends; nominal30mm fan-outs

Approval points / supplier response

Electrical map fixed; supplier drawing and actual service fit pending

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Use exact manufacturer cavity numbering and approved crimp/wire combination.

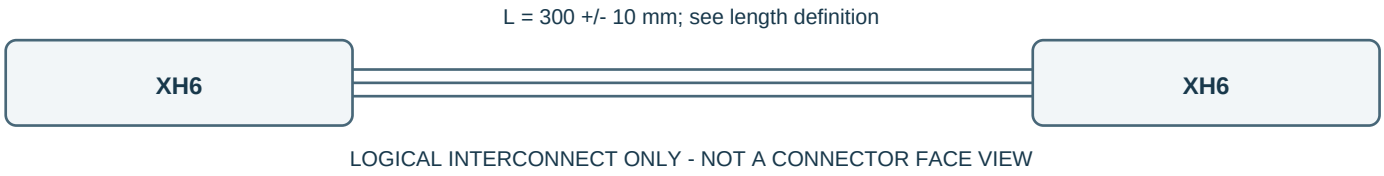
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

H03 | A-B CTRL

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J7 - JST XHP-6; JST SXH-001T-P0.6

End B: B/J7 - JST XHP-6; JST SXH-001T-P0.6



Wire: 26AWG | Existing A-B quotation contract; physical service route pending

A cavity / function	B cavity / function	Signal / purpose	Colour
1	1	GND	black
2	2	PWM0	white
3	3	ENABLE	red
4	4	FAULT_N	yellow
5	5	PWM1 / B pad NC	blue
6	6	DIN0 / B pad NC	green

Jacket and finish

Common insulating flexible jacket; six wires populated and individually labelled; nominal30mm fan-outs

Approval points / supplier response

Positions5/6 must still be wired through, even though PCB B temperature leaves those pads NC; no commoning to GND

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Use exact manufacturer cavity numbering and approved crimp/wire combination.

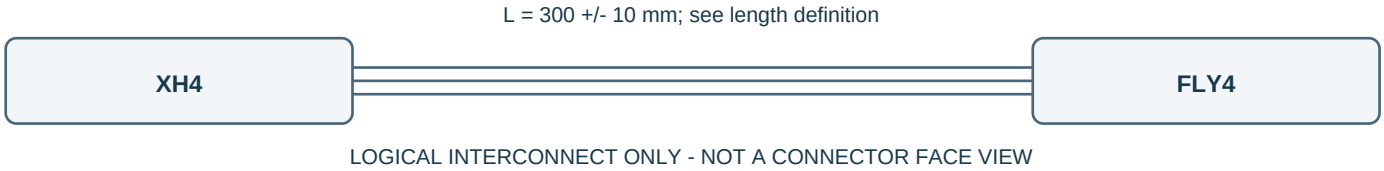
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eXH.pdf

H04 | LCD I2C

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J2 - JST XHP-4; JST SXH-001T-P0.6

End B: Freenove LCD2004 I2C backpack headers - 4 individual Harwin M20-1060100 female1-position housings; 4 Harwin M20-1160042 female crimp contacts



Wire: 26AWG | RFQ proposal only; module/front service slack not measured

A cavity / function	B cavity / function	Signal / purpose	Colour
1	GND	GND	black
2	VCC	+3V3_A	red
3	SDA	I2C_SDA	blue
4	SCL	I2C_SCL	yellow

Jacket and finish

Common insulating jacket; four independent female1-way ends with permanentGND/3V3/SDA/SCL tags, fan-out50mm at module end

Approval points / supplier response

Module physical pin order not assumed: plug each labelled1-way socket on the function-marked male post; 3.3V only in this prototype

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Four independent1-way housings, side-stackable on0.64mm square male posts. Supplier to confirm contact insulation OD/tooling. Label functions rather than assume LCD/GY302 physical order. Proposed documented replacement for generic Dupont, module engagement/clearance still needs real sample check.

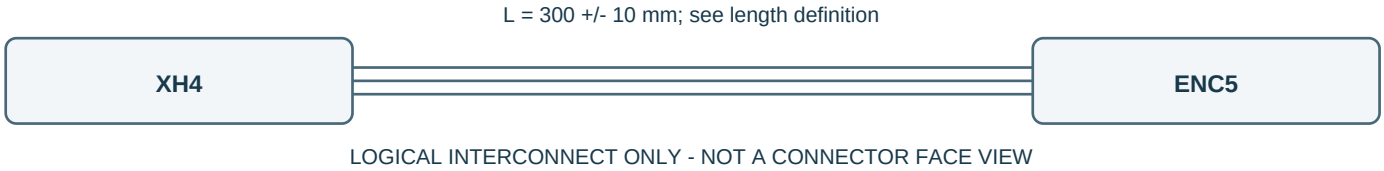
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.harwin.com/products/M20-1060100

H05 | Encoder completed subassembly

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J3 - JST XHP-4; JST SXH-001T-P0.6

End B: ADA377 candidate encoder - Factory-soldered Adafruit377 encoder subassembly / PEC11-4215F-S24 candidate; Encoder contacts A/B/C and two normally-open push-switch contacts SW1/SW2; exact solder view to be confirmed



Wire: 26AWG | RFQ proposal to solder joints; front fit unqualified

A cavity / function	B cavity / function	Signal / purpose	Colour
1	A	ENC_A	white
2	B	ENC_B	blue
3	SW1	ENC_SW	yellow
4	C	GND	black
4	SW2	GND branch	black

Only intentional branch: A/J3.4 GND -> encoderC AND switchSW2. No connection from GND to A/B/SW1 except through normal encoder/switch contacts. Test unpowered state and switch operation, not an indiscriminate insulation test across its contacts.

Jacket and finish

Common insulating jacket; GND branches to encoderC and switchSW2 in factory-insulated splice; separately insulated solder joints, no exposed conductor

Approval points / supplier response

Quote10 fully soldered encoder+harness subassemblies; separate encoder procurement cost and consigned-part alternative. Actual switch variant and terminal orientation need controlled drawing

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Quote procurement+factory soldering as separate cost; alternative customer-consigned encoder price. A free solder-tail harness is not the requested completed assembly. Encoder exact switch variant and terminal view require approval drawing.

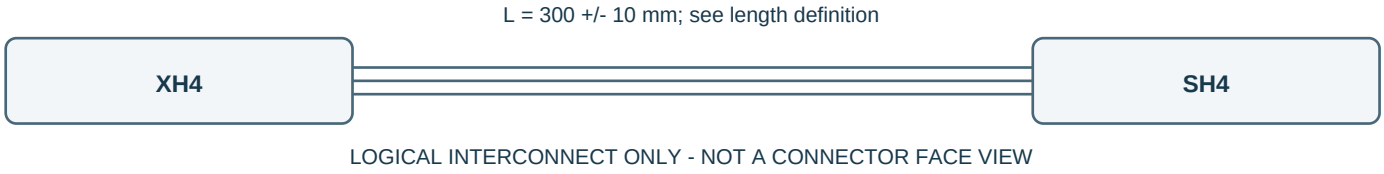
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.gotronic.fr/art-encodeur-24-impulsions-ada377-37481.htm

H06 | TMP117 direct XH-SH option

QUANTITY10 | INTERNAL OPTION | QUOTATION ONLY

End A: A/J4 - JST XHP-4; JST SXH-001T-P0.6

End B: Adafruit4821 STEMMA QT port - JST SHR-04V-S-B; JST SSH-003T-P0.2-H



Wire: 26AWG main, 28AWG short tails | RFQ proposal only; includes short factory transitions

A cavity / function	B cavity / function	Signal / purpose	Colour
1	1	GND	black
2	2	+3V3_A / moduleVIN	red
3	3	I2C_SDA	blue
4	4	I2C_SCL	yellow

Jacket and finish

26AWG main wires withXH-compliant insulationOD0.9..1.9mm;28AWG tails atSH, OD0.4..0.8mm; four factory-made insulated strain-relieved transitions. Main/common jacket plus protected transition, short flexibleSH fan-out20mm nominal; supplier details and dimensions required

Approval points / supplier response

No uniform26AWG inSH and no uniform thick28AWG without approved contact fit. Supplier may propose qualified alternative only with exact contact/wire documentation. This quote replaces two retail adapter cords if approved, no PCB change

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

STEMMA QT compatible receptacle, not JST-PH or XH. 28AWG tails only, factory transition to26AWG main wires.

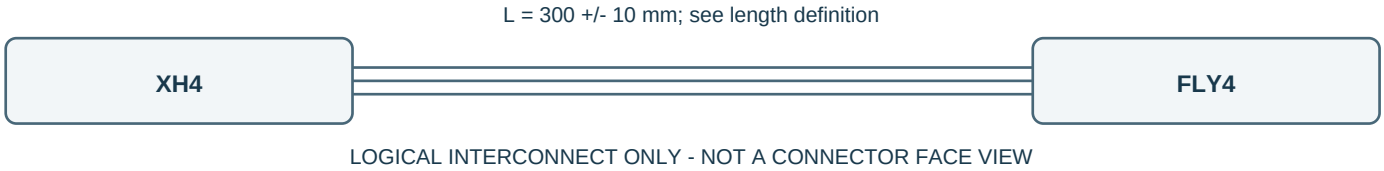
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.jst-mfg.com/product/pdf/eng/eSH.pdf

H07 | BH1750 GY302 I2C

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: A/J5 - JST XHP-4; JST SXH-001T-P0.6

End B: GY302 male headers - 4 individual Harwin M20-1060100 female1-position housings; 4 Harwin M20-1160042 female crimp contacts



Wire: 26AWG | RFQ proposal; optical cassette routing unqualified

A cavity / function	B cavity / function	Signal / purpose	Colour
1	GND	GND	black
2	VCC	+3V3_A	red
3	SDA	I2C_SDA	blue
4	SCL	I2C_SCL	yellow

Jacket and finish

Common insulating flexible jacket; four independent female1-way ends taggedGND/3V3/SDA/SCL;50mm module-end fan-out; no stiff sleeve near cassette

Approval points / supplier response

Actual module male barrette already soldered required;ADDR low0x23 handled on module, not an extra harness conductor; ADA4681 is not a mechanical drop-in substitute

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Four independent1-way housings, side-stackable on0.64mm square male posts. Supplier to confirm contact insulation OD/tooling. Label functions rather than assume LCD/GY302 physical order. Proposed documented replacement for generic Dupont, module engagement/clearance still needs real sample check.

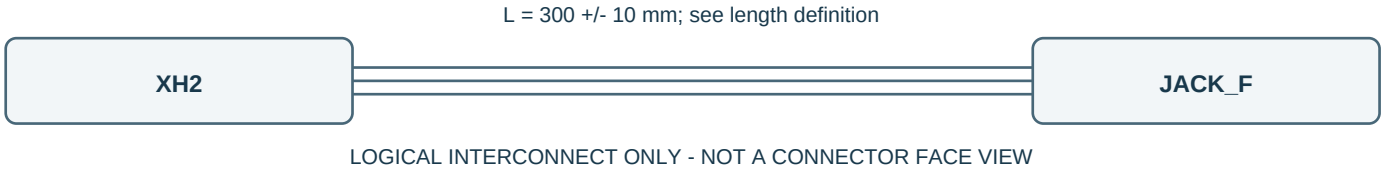
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.harwin.com/products/M20-1060100

H08 | Internal probe panel socket subassembly

QUANTITY10 | INTERNAL | QUOTATION ONLY

End A: B/J4 - JST XHP-2; JST SXH-001T-P0.6

End B: Rear femalePJ392 socket - PJ-3923.5mmTRS panel female socket, TaydaA-2585 dimensional reference; Factory solder to verified tip and sleeve terminals; ring and any switch terminals isolated



Wire: 26AWG | Custom RFQ returns to300mm service proposal; not retailJST1164 length150mm

A cavity / function	B cavity / function	Signal / purpose	Colour
1	TIP	+3V3_B	red
2	SLEEVE	NTC_RAW (NOT GND)	white

TRS RING NOT CONNECTED. Insulate ring and unused switch contacts. SLEEVE = NTC_RAW, NOT GND. Identify physical solder lugs by continuity with a known plug.

Jacket and finish

Common insulating flexible jacket retained behind solder tabs; individual solder-joint insulation, compliant strain relief; include panel socket/nut/washer and factory soldering; labelSLEEVE=NTC_RAW, NOTGND

Approval points / supplier response

Exact socket manufacturer/MPN and lug map by continuity plus approval drawing; no connection to ring, switch lugs, chassis orGND. Check rear-panel removal slack and23mm solder corridor

End reference notes

Use exact manufacturer cavity numbering and approved crimp/wire combination.

Supplier must propose exact manufacturer/MPN/drawing. DAOKIB08XM6DR96 equivalence unqualified. M6x0.5 thread4.5mm, body11.8mm, total22.8mm drawing reference; panel6.3mm hole and local1.5mm web. Include nut/washer, confirm stack. Identify lugs by continuity using knownTRS plug; do not infer lug numbers. Max existing solder corridor23mm depth,10mm diameter to be respected.

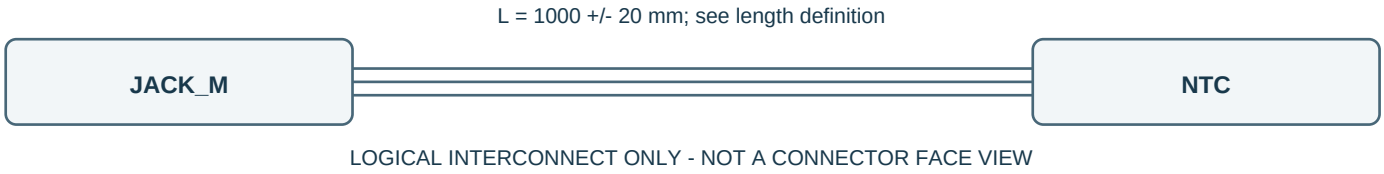
Manufacturer references: https://www.jst-mfg.com/product/pdf/eng/eXH.pdf | https://www.taydaelectronics.com/datasheets/files/PJ-392.pdf

H09 | External jacketed NTC probe option

QUANTITY10 | PROVISIONAL | QUOTATION ONLY

End A: MaleTRS probe plug - Supplier-proposed3.5mmTRS male cable plug with insulated body and real cable clamp; Tip and sleeve only, ring isolated

End B: Passive15k NTC probe - Passive two-wire15kohm NTC probe, R25/certifiedR-T curve target beta~3920; Probe leads1/2 (nonpolar component)



Wire: TBDmm2 / supplier proposal | 1000mm budgetary assumption only, actual application length/rating TBD

A cavity / function	B cavity / function	Signal / purpose	Colour
TIP	NTC1	NTC excitation via internalJ4.1	supplier proposed
SLEEVE	NTC2	NTC measurement via internalJ4.2	supplier proposed

TRS RING NOT CONNECTED. Insulate ring and unused switch contacts. SLEEVE = NTC_RAW, NOT GND. Identify physical solder lugs by continuity with a known plug.

Jacket and finish

Two stranded copper cores inside continuous robust flexible common insulating outer jacket, retained in plug clamp and probe-end strain relief. Supplier declares jacket diameter, temperature rating, medium compatibility and probe seal; individual solder-joint insulation. No bare pair or braid-only jacket

Approval points / supplier response

Budgetary quote10 complete probe+cable+plug assemblies; exactNTC R/T curve, temperature range, probe geometry and plug clamp not selected. Quote proposed cable gauge/MPNs with datasheets; no production release

End reference notes

Exact MPN and mating geometry, body OD and cable-jacket clamp range to propose; the PJ392 document defines mating socket, not a male-plug MPN. Plug must fit proposed robust probe jacket without cutting its outer sheath at clamp.

Exact probe MPN, R25 tolerance, beta reference temperatures/tolerance, temperature range, medium, seal and cable length not approved. Quote candidate only with datasheet; do not substitute10k/100k/DS18B20.

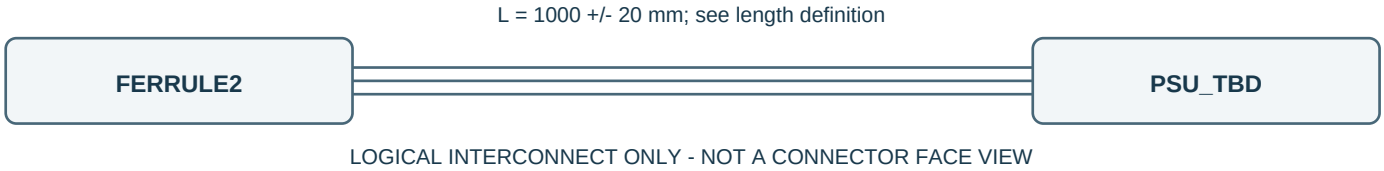
Manufacturer references: https://www.taydaelectronics.com/datasheets/files/PJ-392.pdf | ../system_bom_review.csv

H10 | DC supply input cord

QUANTITY10 | PROVISIONAL | QUOTATION ONLY

End A: B/J2 power terminal - 2 insulated ferrules sized for flexible0.75mm2 copper and Phoenix1715721 MKDS1,5/2-5,08; Supplier proposes exact ferrule MPN/metal length and crimp tooling compatible with terminal

End B: ExternalDC supply output - Power supply output termination TBD; Positive/return functions only



Wire: 0.75mm2 / supplier proposal | 1000mm budgetary assumption; actualPSU/cable path TBD

A cavity / function	B cavity / function	Signal / purpose	Colour
J2.1	DC_POS	VIN_RAW positive	red
J2.2	DC_RETURN	PGND return	black

Jacket and finish

Two stranded copper0.75mm2 cores, common jacketOD3..6.5mm matchingM12 glands; factory-crimped compatible ferrules. Declare temperature rating and min bend radius; no tinned wire under screws

Approval points / supplier response

Nominal12/24V, current,PSU output connector and ferrules TBD. CurrentB engineering reference24V/100W ~4.17A load plus electronics; does not qualify100W at12V. No mains cord in scope

End reference notes

7mm stripping reference from terminal document; do not equate blindly with ferrule tube length. No solder tinning, no terminal block purchase in cable quote.

Nominal12V/24V and actual supply connector not fixed. No assumed barrel, PowerDIN or mains plug.

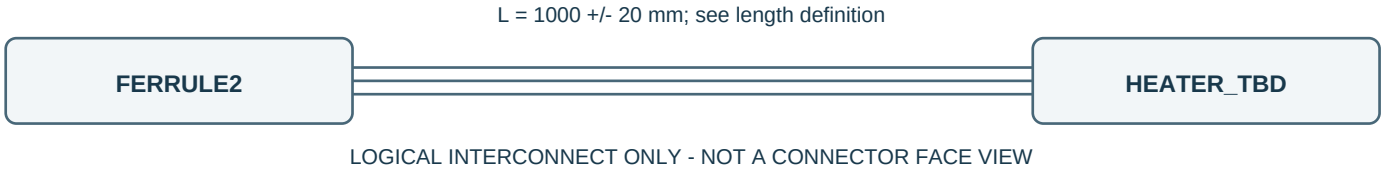
Manufacturer references: https://www.phoenixcontact.com/en-gb/products/printed-circuit-board-terminal-mkds-15-2-508-1715721 | ../system_bom_review.csv

H11 | Heater output cord

QUANTITY10 | PROVISIONAL | QUOTATION ONLY

End A: B/J3 power terminal - 2 insulated ferrules sized for flexible0.75mm2 copper and Phoenix1715721 MKDS1,5/2-5,08; Supplier proposes exact ferrule MPN/metal length and crimp tooling compatible with terminal

End B: Resistive heater - Heater end termination TBD; Positive/switched return functions only



Wire: 0.75mm2 / supplier proposal | 1000mm budgetary assumption; heater application TBD

A cavity / function	B cavity / function	Signal / purpose	Colour
J3.1	HEATER_POS	VIN_PROTECTED positive	red
J3.2	HEATER_RETURN	HEATER_LOW switched return	black

Jacket and finish

Two stranded copper0.75mm2 cores under common jacketOD3..6.5mm; factory-mounted ferrules and appropriate heater-end terminations. Temperature rating at heater end must match actual application, not inferred from ambient internal80degC wire minimum

Approval points / supplier response

Heater and thermal protection arrangement not fixed; no solder-tinned wire under screws. HEATER_LOW is switched power return, not aGND reference; quote current/temperature capability explicitly

End reference notes

7mm stripping reference from terminal document; do not equate blindly with ferrule tube length. No solder tinning, no terminal block purchase in cable quote.

Actual heater leads, insulation temperature and thermal protection topology unresolved.

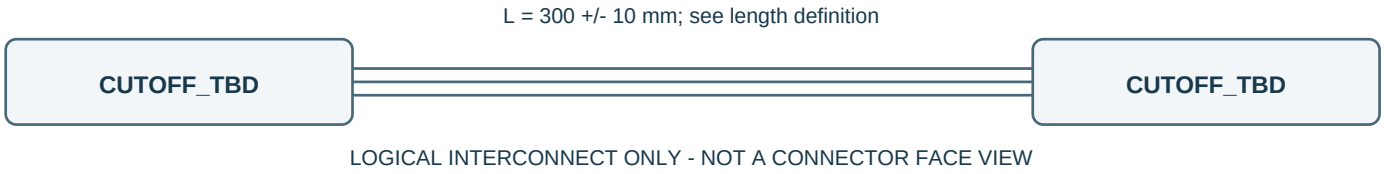
Manufacturer references: https://www.phoenixcontact.com/en-gb/products/printed-circuit-board-terminal-mkds-15-2-508-1715721 | ../system_bom_review.csv

H12 | Independent thermal cutoff lead set

QUANTITY10 | PROVISIONAL | QUOTATION ONLY

End A: Upstream positive heater path - Independent normally-closed thermal protection terminal type TBD; Two series-power terminals only after exact part selected

End B: Cutoff then downstream positive heater path - Independent normally-closed thermal protection terminal type TBD; Two series-power terminals only after exact part selected



Wire: 0.75mm2 / supplier proposal | Budgetary two-lead set, EACH lead300mm; not two redundant power cords

A cavity / function	B cavity / function	Signal / purpose	Colour
UPSTREAM_POS	CUTOFF_TERMINAL_1	Series positive path, location TBD	red
CUTOFF_TERMINAL_2	DOWNSTREAM_POS	Series positive path, location TBD	red

Jacket and finish

10sets, each with two insulated stranded0.75mm2 leads and factory-fitted application terminals. Protective insulation near hot cutoff with supplier-declared rating; terminal sizes cannot be selected before exact protection

Approval points / supplier response

Budgetary functional series topology only. Exact contact ends, trip temperature, DC interruption rating, location and H11 integration TBD. Supplier must avoid double-counting a separateH11 route plus an integrated replacement; price alternative clearly

End reference notes

Trip setting, reset behaviour, location, terminal size and DC interruption rating unresolved. No arbitrary spade/contact size selected.

Trip setting, reset behaviour, location, terminal size and DC interruption rating unresolved. No arbitrary spade/contact size selected.

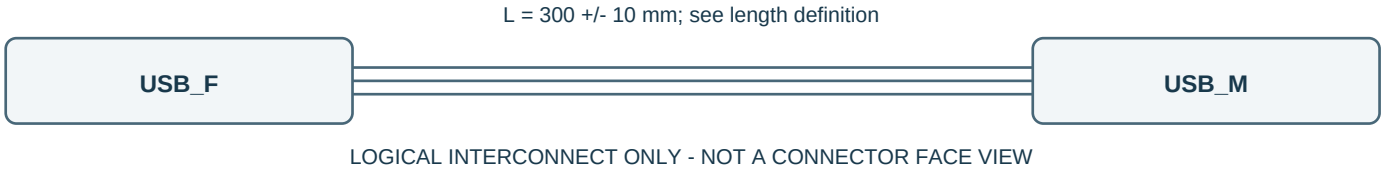
Manufacturer references: ../system_bom_review.csv | ../system_bom_review.csv

H13 | SideUSB-C data extension

QUANTITY10 | PROVISIONAL | QUOTATION ONLY

End A: USB-C female side panel - USB-C female side-panel data extension, supplier proposes controlled MPN; USB2.0 data-capable female interface

End B: USB-C male DevKit - USB-C male data plug to actual ESP32 DevKit; USB2.0 data-capable male interface



Wire: TBDmm2 / supplier proposal | 300mm budgetary cable length excluding plug bodies, supplier datum/geometry required

No discrete USB pin-to-pin table supplied: quote a complete data-capable USB extension. Supplier must document and validate the actual USB-C termination and CC implementation.

Jacket and finish

Complete shielded USB2.0-data cable with female panel mounting and male plug, factory strain relief and common jacket. Supplier-specified USB construction/CC configuration and tests; no discrete flying-pin build

Approval points / supplier response

No compatible16mm panel commercialMPN qualified. Ask exact drawing for hole16/body15/flange22/depth28mm reservation and actualDevKit plug access; any case change requires separate review. External5V+USB operation remains unqualified

End reference notes

Existing reservation: hole16mm, body15mm, flange22mm, internal depth28mm; not a verified commercial part. Exact drawing, male-plug clearance and panel fixings required; M20/M22 is not accepted as drop-in.

Complete USB extension assembly with supplier-validated C configuration/CC wiring, D+/D-,GND,shield/VBUS continuity; not charge-only. No hand-designed universal USB-C pin map. Powered USB+external5V not qualified; service with all other sources disconnected.

Manufacturer references: ../../mechanical/enclosure_parameters.json | ../../devkit_compatibility.md